

Software Testing 5 credits

Syllabus, Master's level, 1DL610

Valid from Autumn 2023

Code

1DL610

Education cycle

Second cycle

Main field(s) of study and in-depth level

Computer Science A1N, Technology A1N

Grading system

Pass with distinction (5), Pass with credit (4), Pass (3), Fail (U)

Finalised by

The Faculty Board of Science and Technology, 18 October 2022

Responsible department

Department of Information Technology

Entry requirements

120 credits including 15 credits in mathematics and 60 credits in computer science/engineering, including a second course in programming in an imperative language and object-oriented programming. Participation in Algorithms and Data Structures I. Proficiency in English equivalent to the Swedish upper secondary course English 6.

Learning outcomes

On completion of the course, the student should be able to:

- describe key techniques and standards in software testing,
- explain and evaluate strategies for software testing for both complete program life cycles and individual phases,
- specify and design test cases and test programs,
- produce appropriate documentation for test management, including test plans, test schedules and test progress monitoring,
- develop code using test driven development,
- be able to use suitable testing tools and discuss their range and capabilities.

Content

Testing methodologies and their practical application in software development. Different aspects of testing: Black box testing where testing is done without knowledge of how the program is written; white box testing where the developer

tries to guarantee that every statement, execution path and method is executed during the testing and finally unit testing which is a practical design methodology where test cases are developed as each function or method is written.

Instruction

Lectures, laboratory exercises.

Assessment

Written exam. Oral and written assessment of exercises.

If there are special reasons for doing so, an examiner may make an exception from the method of assessment indicated and allow a student to be assessed by another method. An example of special reasons might be a certificate regarding special pedagogical support from the disability coordinator of the university.

Other regulations

This course cannot be included in a degree together with Software testing and maintenance DV2 (1DL600), Testing Methodology (1DT053).